

Auteur: Anna Voronovska
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 44.7

$$\int \frac{x^2 - 43}{x^3 + x + 10} dx$$

Noemer ontbinden, $x = -2$:

$$(-2)^3 + (-2) + 10 = -8 - 2 + 10 = 0$$

Dus $(x + 2)$ is een factor:

Horner deling met $x = -2$:

$$\begin{array}{r|rrrr} & 1 & 0 & 1 & 10 \\ -2 & & -2 & 4 & -10 \\ \hline & 1 & -2 & 5 & 0 \end{array}$$

$$\text{Dus } x^3 + x + 10 = (x + 2)(x^2 - 2x + 5)$$

$$x^3 + x + 10 = (x + 2)(x^2 - 2x + 5)$$

Partieelbreuken:

$$\frac{x^2 - 43}{(x + 2)(x^2 - 2x + 5)} = \frac{A}{x + 2} + \frac{Bx + C}{x^2 - 2x + 5}$$

$$x^2 - 43 = A(x^2 - 2x + 5) + (Bx + C)(x + 2)$$

$$A = \frac{x^2 - 43}{x^2 - 2x + 5} \Big|_{x=-2} = \frac{4 - 43}{4 + 4 + 5} = \frac{-39}{13} = -3$$

$$B(1 + 2i) + C \Rightarrow \frac{x^2 - 43}{x + 2} \Big|_{x=1+2i} = \frac{(1 + 2i)^2 - 43}{(1 + 2i) + 2} = \frac{1 + 4i - 4 - 43}{3 + 2i} = \frac{-46 + 4i}{3 + 2i}$$

$$= \frac{(-46 + 4i)(3 - 2i)}{(3 + 2i)(3 - 2i)} = \frac{-138 + 92i + 12i + 8}{13} = \frac{-130 + 104i}{13} = -10 + 8i$$

$$B(1 + 2i) + C = -10 + 8i$$

$$\Rightarrow 2B = 8$$

$$\Rightarrow B = 4$$

$$B + C = -10$$

$$\Rightarrow C = -14$$

Dus:

$$\frac{x^2 - 43}{(x + 2)(x^2 - 2x + 5)} = \frac{-3}{x + 2} + \frac{4x - 14}{x^2 - 2x + 5}$$

Integreren:

Eerste integraal:

$$\int \frac{-3}{x + 2} dx = -3 \ln |x + 2|$$

Tweede integraal:

$$\int \frac{4x - 14}{x^2 - 2x + 5} dx = \int \frac{4x - 4}{x^2 - 2x + 5} dx - \int \frac{10}{(x - 1)^2 + 4} dx$$

Eerste deel: $u = x^2 - 2x + 5 \Rightarrow du = (2x - 2)dx$:

$$\int \frac{4x - 4}{x^2 - 2x + 5} dx = 2 \int \frac{du}{u} = 2 \ln |x^2 - 2x + 5|$$

Tweede deel: $t = x - 1$:

$$\int \frac{10}{(x - 1)^2 + 4} dx = \frac{10}{2} \operatorname{Bgtan} \left(\frac{x - 1}{2} \right) = 5 \operatorname{Bgtan} \left(\frac{x - 1}{2} \right)$$

Alles samen:

$$\int \frac{x^2 - 43}{x^3 + x + 10} dx = -3 \ln |x + 2| + 2 \ln(x^2 - 2x + 5) - 5 \operatorname{Bgtan} \left(\frac{x - 1}{2} \right) + c$$

References